Theory of Valuations

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Introduction. The valuation of a field/ring is a way to study abstract structures using analysis methods. For example, if we have a valued field/ring we can talk about completeness or study sequences. Also, in this paper I will talk about the p-adic numbers for p a prime, which we can think about as a base-p number system (binary when $p = 2$). Also, we can ask interesting questions in the field of p-adic numbers like: is $\sqrt{2}$ a p-adic number? i.e. for example, does the equation $x^2 - 2 = 0$ have a solution in the 7-adic or the 5-adic numbers?

Abstract. In this paper I mostly used the book Algebraic Number Theory by Neukirch [1]. I talked about the p-adic numbers, gave some examples and wrote two codes: one code to convert a rational number into its p-adic representation, and the second code to check if a number has a square root in the p-adic numbers for a certain p. In the next sections I talked about the theory of valuations, the p-adic valuation, the field of rational functions and its completion with respect to a nonarchimedean valuation (formal Laurent series), and Puiseux series.

1 p-adic numbers

Let p be a prime number. The *p-adic integers* $\mathbb{Z}_p$ is the set with the elements in the form

$$\mathbb{Z}_p = \left\{ \sum_{i=0}^{\infty} a_i p^i : 0 \leq a_i < p \right\}. \tag{1}$$

$\mathbb{Z}_p$ with two operations, addition and multiplication (with carrying, as for numbers written in base p), forms a ring. And its field of fractions is the set of *p-adic numbers* $\mathbb{Q}_p$:

$$\mathbb{Q}_p = \left\{ \sum_{i=m}^{\infty} a_i p^i : m \in \mathbb{Z},\ 0 \leq a_i < p \right\}. \tag{2}$$

We can write the elements in $\mathbb{Q}_p$ by the coefficients of the p powers as $a_m a_{m+1} a_{m+2} \cdots$, and we use a comma after the a_0 coefficient (the coefficient with no p power). There is a 1-1 map (embedding) from the set of rational numbers $\mathbb{Q}$ into $\mathbb{Q}_p$: a rational number corresponds to the element of $\mathbb{Q}_p$ whose sum in (2) has that value, when we compute the sum without concern about the convergence (as in the next example).

> **Example 1.1**. Find the rational number which corresponds to the 7-adic number

$3,545454\cdots$

$$\begin{aligned}3,54545454\cdots &= 3+5\cdot 7+4\cdot 7^2+5\cdot 7^3+4\cdot 7^4+\cdots\\ &= 3+(5\cdot 7+4\cdot 7^2)(1+7^2+7^4+\cdots)\\ &= 3+(5\cdot 7+4\cdot 7^2)\left(\frac{1}{1-7^2}\right)\\ &= 3+\frac{35+196}{-48}=\frac{-144+35+196}{-48}=\frac{-87}{48}=\frac{-29}{16}.\end{aligned}$$

Also, if we want to convert from the rationals to the p-adics, we do that coefficient by coefficient, in the following way.

Let $c \in \mathbb{Q}$. Then we factor the p powers from c, so we have $c = p^m b_m$ where $p \nmid b_m$ (that is, p divides neither the numerator nor the denominator of b_m). Now for b_m we find an integer $0 \le a_m < p$ such that $p \mid b_m - a_m$, and we write $b_m = a_m + p \cdot b_{m+1}$. This a_m is the coefficient of p^m. Then we do the same for b_{m+1} to find a_{m+1}, the coefficient of p^{m+1}, and we keep going.

Example 1.2. What does $\frac{110}{7}$ correspond to in the 11-adics? First we factor the 11 powers:

$$\frac{110}{7} = 11 \cdot \frac{10}{7}.$$

Here we will begin with finding a_1, because the fraction has p^1 as a factor. So we want to find $0 \le a_1 < 11$ such that $\frac{10}{7} - a_1$ has 11 as a factor. The solution is 3; to see this,

$$\frac{10}{7} - 3 = \frac{10-21}{7} = 11 \cdot \frac{-1}{7}.$$

So $a_1 = 3$. Now we want to find a_2 such that $\frac{-1}{7} - a_2$ has 11 as a factor; again it will be 3:

$$\frac{-1}{7} - 3 = \frac{-1-21}{7} = 11 \cdot \frac{-2}{7}.$$

So $a_2 = 3$. Now we want to find a_3 such that $\frac{-2}{7} - a_3$ has 11 as a factor. The answer is 6:

$$\frac{-2}{7} - 6 = \frac{-44}{7} = 11 \cdot \frac{-4}{7}.$$

So $a_3 = 6$. And then $a_4 = 1$:

$$\frac{-4}{7} - 1 = \frac{-4-7}{7} = 11 \cdot \frac{-1}{7}.$$

Then it will be periodic, because we get $\frac{-1}{7}$, which we already had. Hence

$$\frac{110}{7} = 0,3\,3\,6\,1\,3\,6\,1\cdots$$

We can check that it is correct in two ways. First, we use the geometric sum:

$$0,3361361\cdots = 3\cdot 11+(3\cdot 11^2+6\cdot 11^3+11^4)(1+11^3+\cdots) = 33-\frac{3\cdot 11^2+6\cdot 11^3+11^4}{11^3-1} = \frac{110}{7}.$$

Second, if we multiply $\frac{110}{7}$ by 7 we get 110, which as an 11-adic is $0, 10$ (here 10 is one coefficient, since $110 = 10 \cdot 11$):

$$7\,(0, 3361361\cdots) = 0, 10\ 0\ 0\ 0\cdots$$

Multiplying 7 with 3 we get $21 = 10 + 1 \cdot 11$, so the carried number is 1; then $7 \cdot 3 + 1 = 22 = 0 + 2 \cdot 11$, so the carried number is 2; then $7 \cdot 6 + 2 = 44 = 0 + 4 \cdot 11$, so the carried number is 4; then $7 \cdot 1 + 4 = 11 = 0 + 1 \cdot 11$, and so on.

There is an interesting exercise in [1, Exercise 1, p. 106]:

Proposition 1.3. *A p-adic number $a = \sum_{i=m}^{\infty} a_i p^i$ is a rational number if and only if the sequence of coefficients $a_m, a_{m+1}, \cdots$ is periodic after a finite number of terms.*

Proof. ($\Rightarrow$) Suppose that $a \in \mathbb{Q}_p$ is a rational number, say $\frac{s}{t}$. We want to show the p-adic representation will be periodic after a finite number of terms. First, we factor the p powers so that

$$\frac{s}{t} = p^m \cdot \frac{s'}{t'}, \quad \text{with } p \nmid s't' \text{ and } t' > 0.$$

We use Euclid division $s' = q \cdot t' + r$ with $q \in \mathbb{Z}$ and $-t' \leq r \leq -1$ (we choose q so that the remainder is negative), so we have $\frac{s'}{t'} = q + \frac{r}{t'}$. Since t' is relatively prime with p, we can use Euler's theorem (the generalization of Fermat's little theorem) to get

$$p^{\varphi(t')} \equiv 1 \pmod{t'},$$

where $\varphi(t')$ is the cardinality of the set $A = \{x : 1 \leq x \leq t' \ \&\ \gcd(x, t') = 1\}$. Thus $p^{\varphi(t')} - 1 = kt'$ with $k \geq 1$. Let $\varphi(t') = N$ and let $x = -kr$. Then

$$\frac{r}{t'} = \frac{kr}{kt'} = \frac{-x}{p^N - 1} = \frac{x}{1 - p^N} = x \underbrace{\left(1 + p^N + p^{2N} + \cdots\right)}_{1/(1-p^N)}.$$

Since $x \in \mathbb{N}$, we write its p-adic expansion, and $a_N = a_{N+1} = \cdots = 0$ in its expansion because $-t' \leq r \leq -1 \Rightarrow 0 < x = -kr \leq kt' = p^N - 1 < p^N$. Suppose $x = a_0 + a_1 p + \cdots + a_{N-1} p^{N-1}$. Then we get that

$$\frac{r}{t'} = a_0 + a_1 p + \cdots + a_{N-1} p^{N-1} + a_0 p^N + a_1 p^{1+N} + \cdots + a_{N-1} p^{2N-1} + \cdots$$

which is periodic. Thus

$$\frac{s}{t} = p^m \cdot \left(q + a_0 + a_1 p + \cdots + a_{N-1} p^{N-1} + a_0 p^N + a_1 p^{1+N} + \cdots\right).$$

Multiplying by p^m only shifts the coefficients. I will show that adding an integer does not affect the periodicity of a p-adic number. Since any integer q is a sum of 1's or of -1's, if adding 1 and adding -1 do not affect the periodicity of a p-adic number, then adding an integer will not. Let $a_0 + a_1 p + \cdots$ be the p-adic integer. If $a_0 < p - 1$, then adding 1 will just affect the

first coefficient. If $a_0 = p-1$, then a_0 becomes 0 and a 1 will carry on to the next coefficient, and so on. So adding 1 changes only finitely many coefficients, unless all coefficients from some point on are $p-1$; then those all become 0. In both cases the result is periodic after a finite number of terms. Adding -1 is the same with borrowing instead of carrying: it changes only finitely many coefficients, unless all coefficients from some point on are 0, and then those all become $p-1$.

($\Leftarrow$) Let $a = \sum_{i=m}^{\infty} a_i p^i$; multiplying by p^{-m} we can assume $m = 0$. Suppose it is periodic after N terms, and suppose the length of the string which repeats is k. Then

$$\begin{aligned} a &= a_0 + \cdots + a_{N-1}p^{N-1} + (a_N p^N + \cdots + a_{N+k-1}p^{N+k-1})(1 + p^k + p^{2k} + \cdots) \\ &= a_0 + \cdots + a_{N-1}p^{N-1} + \frac{a_N p^N + \cdots + a_{N+k-1}p^{N+k-1}}{1-p^k}, \end{aligned}$$

which is a rational number. □

Another presentation of the p-adics. There is another way to write the p-adic integers: instead of writing the coefficients $a_0, a_1 a_2 \cdots$ we can write the partial sums of $\sum_{i=0}^{\infty} a_i p^i$. We define

$$\begin{aligned} s_0 &= a_0, \\ s_1 &= a_0 + a_1 \cdot p, \\ s_2 &= a_0 + a_1 \cdot p + a_2 \cdot p^2. \end{aligned}$$

In general it will be

$$s_n = \sum_{i=0}^{n} a_i p^i.$$

Define the **inverse (projective) limit** of the rings $\mathbb{Z}/p^n\mathbb{Z}$ to be the set

$$\varprojlim \mathbb{Z}/p^n\mathbb{Z} = \left\{ \prod_{i=0}^{\infty}(s_i) : s_i \in \mathbb{Z}/p^{i+1}\mathbb{Z} \;\&\; s_{i+1} \equiv s_i \pmod{p^{i+1}} \right\}.$$

Now we need to show that the two representations of the p-adics are the same. We define the map

$$\phi : \mathbb{Z}_p \to \varprojlim \mathbb{Z}/p^n\mathbb{Z} \quad \text{by} \quad \phi(a_0, a_1 a_2 \cdots) = \prod_{j=0}^{\infty} \left(\sum_{i=0}^{j} a_i p^i \right).$$

It is a bijection, with the inverse

$$\phi^{-1}\left(\prod_{i=0}^{\infty}(s_i) \right) = \left(s_0, \frac{s_1 - s_0}{p}, \frac{s_2 - s_1}{p^2}, \frac{s_3 - s_2}{p^3}, \cdots \right),$$

where each s_i is taken as an integer with $0 \le s_i < p^{i+1}$. And this representation is useful in proving the following proposition.

Proposition 1.4. *Let $Q(x)$ be a polynomial with integer coefficients. Then the equation $Q(x) = 0$ has a solution in $\mathbb{Z}_p$ iff $Q(x) \equiv 0 \pmod{p^n}$ has a solution for every n.*

Proof. ($\Rightarrow$) Suppose that the equation $Q(x) = 0$ has a solution in the p-adics, i.e. there is $x_p \in \mathbb{Z}_p$ such that $Q(x_p) = 0$. Then, writing s_{n-1} for the partial sum of x_p up to p^{n-1}, we have $x_p \equiv s_{n-1} \pmod{p^n}$, so clearly from the mod properties $Q(s_{n-1}) \equiv Q(x_p) = 0 \pmod{p^n}$ for any $n \in \mathbb{N}$.

($\Leftarrow$) Now suppose that $Q(x) \equiv 0 \pmod{p^n}$ has a solution for any $n \in \mathbb{N}$. Note that a solution mod p^N is also a solution mod p^n for every $n \leq N$. Call a solution t mod p^n *good* if for every $N \geq n$ there is a solution mod p^N which is $\equiv t \pmod{p^n}$.

First, there is a good solution mod p. Indeed, every solution mod p^N reduces mod p to one of the finitely many elements of $\mathbb{Z}/p\mathbb{Z}$, so one element s_0 is the reduction of solutions mod p^N for infinitely many N, and hence for all N (a solution mod a higher power reduces to a solution mod a lower power).

Next, if s_n is a good solution mod p^{n+1}, then one of its p lifts $s_n + c\,p^{n+1}$, $0 \leq c < p$, is a good solution mod p^{n+2}, by the same argument: the solutions mod p^N ($N \geq n+2$) which are $\equiv s_n \pmod{p^{n+1}}$ reduce mod p^{n+2} to one of these p lifts, so one of them is used for infinitely many N, and hence for all N. Call it s_{n+1}.

So we get $s_0, s_1, s_2, \cdots$ with $s_i \in \mathbb{Z}/p^{i+1}\mathbb{Z}$, $s_{i+1} \equiv s_i \pmod{p^{i+1}}$ and $Q(s_i) \equiv 0 \pmod{p^{i+1}}$. This is an element of $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$, so by the bijection ϕ it is a p-adic integer $x_p = \phi^{-1}(\prod(s_i))$. Then $Q(x_p) \equiv Q(s_i) \equiv 0 \pmod{p^{i+1}}$ for every i, i.e. all the coefficients of $Q(x_p)$ are 0, so $Q(x_p) = 0$. □

Example 1.5. We will show that $\sqrt{2}$ is in the 7-adics but not in the 11-adics. To prove that $\sqrt{2}$ is in the 7-adics, it is enough, by the previous proposition, to show there is a solution to the following congruence:

$$x^2 \equiv 2 \pmod{7^n}, \quad \forall n \in \mathbb{N}.$$

I will use induction. For $n = 1$ it is easy to see that $3^2 \equiv 2 \pmod{7}$. Now suppose that

$$x^2 \equiv 2 \pmod{7^n}$$

is solvable, and let one of the solutions be

$$x_n = \sum_{i=0}^{n-1} a_i \cdot 7^i, \qquad a_0 = 3.$$

Now I will show there is $a_n \in \{0, 1, 2, \cdots, 6\}$ such that

$$\underbrace{(x_n + a_n \cdot 7^n)^2}_{x_{n+1}^2} \equiv 2 \pmod{7^{n+1}}.$$

We know that $x_n^2 \equiv 2 \pmod{7^n}$, which means that $7^n \mid x_n^2 - 2$. I will use that to solve the congruence:

$$\begin{aligned}
(x_n + a_n \cdot 7^n)^2 &\equiv 2 \pmod{7^{n+1}} \\
x_n^2 - 2 + 2x_n a_n \cdot 7^n + a_n^2\, 7^{2n} &\equiv 0 \pmod{7^{n+1}} \\
7^n \left(\frac{x_n^2 - 2}{7^n} + 2x_n a_n \right) &\equiv 0 \pmod{7^{n+1}}
\end{aligned}$$

(the red term is divisible by 7^{n+1}). Thus

$$\frac{x_n^2-2}{7^n}+2x_na_n\equiv 0 \pmod 7.$$

We will simplify it (all congruences below are mod 7):

$$\begin{aligned}
\frac{x_n^2-2}{7^n}+2x_na_n &\equiv 0 && (1)\\
2x_na_n &\equiv -\frac{x_n^2-2}{7^n} && (2)\\
6\cdot a_n &\equiv 6\cdot\frac{x_n^2-2}{7^n} && (3)\\
a_n &\equiv \frac{x_n^2-2}{7^n} && (4)\\
a_n &\equiv \frac{(x_{n-1}+a_{n-1}\cdot 7^{n-1})^2-2}{7^n} && (5)\\
a_n &\equiv \frac{x_{n-1}^2+2a_{n-1}x_{n-1}\cdot 7^{n-1}+a_{n-1}^2\cdot 7^{2n-2}-2}{7^n} && (6)\\
a_n &\equiv \frac{x_{n-1}^2-2+2a_{n-1}x_{n-1}\cdot 7^{n-1}}{7^n}+a_{n-1}^2\cdot 7^{n-2} && (7)\\
a_n &\equiv \frac{x_{n-1}^2-2+2a_{n-1}x_{n-1}\cdot 7^{n-1}}{7^n} && (8)
\end{aligned}$$

Here are the details of what I did:

- from (2) to (3) I used the fact that $x_n\equiv 3 \pmod 7$, so $2x_n\equiv 6 \pmod 7$, and $-1\equiv 6 \pmod 7$;
- from (3) to (4) I multiplied by 6, because it is its own inverse: $6\cdot 6=36\equiv 1 \pmod 7$;
- from (4) to (5) using that $x_n=x_{n-1}+a_{n-1}\cdot 7^{n-1}$;
- from (7) to (8): since we are taking mod 7, when $n\geq 3$ we get $a_{n-1}^2\cdot 7^{n-2}\equiv 0$.

Let's compute the first four terms. We know that $a_0=3$, and to find a_1 I will use (4):

$$a_1\equiv\frac{3^2-2}{7}\equiv 1.$$

Now to find a_2 I will use (4) again, with $x_2=3+7=10$:

$$a_2\equiv\frac{10^2-2}{7^2}\equiv 2.$$

We have $x_3=3+7+2\cdot 7^2$; to find a_3 I will use (8):

$$a_3\equiv\frac{x_2^2-2+2a_2x_2\cdot 7^2}{7^3}\equiv\frac{10^2-2+2(2)(10)\cdot 7^2}{7^3}=\frac{2058}{343}=6.$$

So $x_4=3+7+2\cdot 7^2+6\cdot 7^3$.

To show that $\sqrt{2}$ is not in the 11-adics, we look at the elements of $\mathbb{Z}/11\mathbb{Z}$, which are $\{0,1,2,\cdots,10\}$; their squares are $\{0,1,4,9,5,3\}$. Hence

$$x^2 \equiv 2 \pmod{11}$$

has no solution, so $\sqrt{2}$ is not an element of the 11-adic integers. (It is not in $\mathbb{Q}_{11}$ either: if $x = 11^m u$ with $u \in \mathbb{Z}_{11}$, $11 \nmid u$, and $x^2 = 2$, then $11^{2m}u^2 = 2$ forces $m = 0$.)

What I did to show that $\sqrt{2}$ is in the 7-adics and to evaluate the first 4 terms has a generalization in the following lemma.

Hensel's Lemma (first version). *If $Q(x)$ is a polynomial with integer coefficients and $Q(x) \equiv 0 \pmod{p^n}$ has a solution, say $t \in \mathbb{Z}/p^n\mathbb{Z}$, then this solution can be lifted to a solution $\pmod{p^{n+1}}$ if $Q'(t) \not\equiv 0 \pmod{p}$.*

Proof. Suppose $Q(x) = \sum_{i=0}^{m} a_i x^i$ and suppose

$$Q(t) = \sum_{i=0}^{m} a_i t^i \equiv 0 \pmod{p^n}.$$

We need to look for $0 \le c < p$ such that $t + cp^n$ satisfies $Q(t + cp^n) \equiv 0 \pmod{p^{n+1}}$. Then

$$\begin{aligned}
\sum_{i=0}^{m} a_i (t + cp^n)^i &\equiv 0 \pmod{p^{n+1}} \\
a_0 + \sum_{i=1}^{m} a_i \Big(t^i + it^{i-1}(cp^n) + \underbrace{\tfrac{i(i-1)}{2} t^{i-2}(cp^n)^2 + \cdots + (cp^n)^i}_{\equiv 0 \pmod{p^{n+1}}} \Big) &\equiv 0 \pmod{p^{n+1}} \\
a_0 + \sum_{i=1}^{m} a_i \big(t^i + it^{i-1}(cp^n) \big) &\equiv 0 \pmod{p^{n+1}} \\
Q(t) + cp^n Q'(t) &\equiv 0 \pmod{p^{n+1}}.
\end{aligned}$$

Dividing by p^n, this is $\frac{Q(t)}{p^n} + c\,Q'(t) \equiv 0 \pmod{p}$, and this last congruence has a solution c if $Q'(t) \not\equiv 0 \pmod{p}$, namely $c \equiv -\frac{Q(t)}{p^n} Q'(t)^{-1} \pmod{p}$. □

2 Valuations

A *valuation* on a field $\mathbb{F}$ is a function $|\cdot| : \mathbb{F} \to \mathbb{R}$ satisfying

- $|x| \ge 0$ for any $x \in \mathbb{F}$,
- $|x| = 0$ iff $x = 0$,
- $|xy| = |x||y|$,
- $|x + y| \le |x| + |y|$.

From the definition of the valuation, if 1 is the multiplicative identity of $\mathbb{F}$, then $|1| \cdot |1| = |1 \cdot 1| = |1|$, so $|1| = 1$.

A valuation is called *nonarchimedean* if there is $M \in \mathbb{R}$ such that $|n| \leq M$ for all $n \in \mathbb{N}$; otherwise it is called *archimedean*.

Proposition 2.1. *A valuation is nonarchimedean iff it satisfies the strong triangle inequality, i.e.* $|x+y| \leq \max\{|x|, |y|\}$.

Proof. ($\Rightarrow$) Suppose that a valuation $|\cdot|$ on the field $\mathbb{F}$ is nonarchimedean, i.e. there is $M \in \mathbb{R}$ such that $|n| \leq M$ for all $n \in \mathbb{N}$. Let $x, y \in \mathbb{F}$ be nonzero elements and suppose that $\max\{|x|, |y|\} = |y|$. Then

$$\begin{aligned} |x+y|^n &= |(x+y)^n| \\ &= \left| \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i} \right| \\ &\leq \sum_{i=0}^{n} \left| \binom{n}{i} \right| |y|^n \\ &\leq M\,|y|^n (n+1). \end{aligned}$$

Hence $|x+y| \leq M^{\frac{1}{n}} |y| (n+1)^{\frac{1}{n}}$; if $n \to \infty$ then we get $|x+y| \leq |y| = \max\{|x|, |y|\}$.

($\Leftarrow$) Suppose that a valuation on a field $\mathbb{F}$ satisfies $|x+y| \leq \max\{|x|, |y|\}$ for any $x, y \in \mathbb{F}$. Let $n \in \mathbb{N}$; then

$$|n| = |(n-1)+1| \leq \max\{|n-1|, |1|\}.$$

Thus, by induction, $|n| \leq |1|$ for any $n \in \mathbb{N}$, so we set $M = |1| = 1$. □

Example 2.2. We can define a nonarchimedean valuation on $\mathbb{Q}$. Let p be a prime. If $x \in \mathbb{Q}$, $x \neq 0$, is such that $x = p^m \cdot \frac{a}{b}$ with $p \nmid ab$, then we define the map $v_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ as $v_p(x) = m$. Also, we let $v_p(0) = \infty$.
This v_p is called the *exponential valuation*, and it satisfies the following properties:

- $v_p(x) = \infty$ iff $x = 0$,
- $v_p(xy) = v_p(x) + v_p(y)$,
- $v_p(x+y) \geq \min\{v_p(x), v_p(y)\}$.

To prove the second, suppose $x = p^m u$, $y = p^n w$ where p divides no numerator or denominator of u, w; then $xy = p^{m+n} uw$, so clearly $v_p(xy) = m + n = v_p(x) + v_p(y)$.
To prove the third property, let again $x = p^m u$, $y = p^n w$, and suppose that $m < n$, i.e. $v_p(x) < v_p(y)$. Then

$$v_p(x+y) = v_p\big(p^m(u + p^{n-m} w)\big) = m \geq \min\{v_p(x), v_p(y)\}.$$

If $m = n$, then $x + y = p^m(u+w)$, and $u + w$ has no p in its denominator, so $v_p(x+y) \geq m = \min\{v_p(x), v_p(y)\}$. (If x or y is 0 the property is clear.)
Then we define the nonarchimedean valuation $|\cdot|_p : \mathbb{Q} \to \mathbb{R}$ by

$$|x|_p = p^{-v_p(x)} \qquad (\text{with } |0|_p = p^{-\infty} = 0).$$

From this definition we can observe the following properties:

- $|x|_p \geq 0$ for all $x \in \mathbb{Q}$,
- $|x|_p = 0$ iff $x = 0$,
- $|xy|_p = |x|_p|y|_p$,
- $|x+y|_p \leq \max\{|x|_p, |y|_p\}$.

The first is clear, since $|x|_p = p^{-v_p(x)}$, which is always positive or 0. The second property is true from the fact $\lim_{t\to\infty} p^{-t} = 0$, and $v_p(x) = \infty$ only for $x = 0$. The third: $|xy|_p = p^{-v_p(xy)} = p^{-v_p(x)-v_p(y)} = p^{-v_p(x)}p^{-v_p(y)} = |x|_p|y|_p$. The fourth one: $|x+y|_p = p^{-v_p(x+y)} \leq p^{-\min\{v_p(x),v_p(y)\}} = \max\{|x|_p, |y|_p\}$. So $|\cdot|_p$ defines a valuation on $\mathbb{Q}$.

Example 2.3. Consider the 3-adic valuation on $\mathbb{Q}$. For any integer $n \in \mathbb{Z}$ we get that $|n|_3 \leq 1$. Let $A = \{\frac{a}{b} : 1 < |\frac{a}{b}|_3 < 4\}$; this set contains all rational numbers $\frac{a}{b}$ (in lowest terms) such that $3 \mid b$ but $3^2 \nmid b$, since these are exactly the ones with $|\frac{a}{b}|_3 = 3$. The 3-adic valuation defines a metric on the rational numbers; for instance $d(\frac{1}{3}, 10) = |\frac{1}{3} - 10|_3 = |\frac{-29}{3}|_3 = 3$. Also, for any $a, b \in \mathbb{Z}$ with $3 \nmid b$ we have $|\frac{1}{3} - \frac{a}{b}|_3 = 3$. Suppose $b = 3^r c$ where $r \geq 2$, $3 \nmid c$ and $3 \nmid a$. Then

$$d\left(\frac{1}{3}, \frac{a}{b}\right) = \left|\frac{1}{3} - \frac{a}{b}\right|_3 = \left|\frac{3^{r-1}c - a}{3^r c}\right|_3 = 3^r,$$

since $3 \nmid 3^{r-1}c - a$. In general, suppose $r_1, r_2 \in \mathbb{Q}$ and $r_1 = p^{e_1}q_1$, $r_2 = p^{e_2}q_2$ with $v_p(q_1) = v_p(q_2) = 0$ and $e_1 < e_2$. Then

$$d_p(r_1, r_2) = |p^{e_1}q_1 - p^{e_2}q_2|_p = |p^{e_1}(q_1 - p^{e_2-e_1}q_2)|_p = p^{-e_1} = |r_1|_p.$$

Proposition 2.4. *For any nonzero rational number $a \in \mathbb{Q}$ we have*

$$\prod_{p \text{ prime}} |a|_p = \frac{1}{\|a\|},$$

where $\|a\|$ is the absolute value of a.

Proof. We write the factorization of a as $a = \text{sign}(a)\, p_1^{e_1} p_2^{e_2} \cdots p_n^{e_n}$ with $e_i \in \mathbb{Z}$; then $|a|_{p_i} = \frac{1}{p_i^{e_i}}$, and $|a|_p = 1$ for every other prime p. Thus

$$\prod_{p \text{ prime}} |a|_p = \frac{1}{p_1^{e_1} p_2^{e_2} \cdots p_n^{e_n}} = \frac{1}{\|a\|}.$$

□

We say two valuations $|\cdot|_a$, $|\cdot|_b$ on $\mathbb{F}$ are *equivalent* if there exists $s \in \mathbb{R}^+$ such that $|x|_a = |x|_b^s$ for all $x \in \mathbb{F}$.

Proposition 2.5. *Two valuations are equivalent if and only if they induce the same topology. (The following is a recap of what is in [2].)*

Proof. Let $|\cdot|_1, |\cdot|_2$ be two valuations on the same field $\mathbb{K}$. Suppose the two valuations are equivalent, i.e. there is $s \in \mathbb{R}^+$ such that $|x|_1 = |x|_2^s$ for all $x \in \mathbb{K}$. Then the balls

$$B_1(a, r^s) = \{x \in \mathbb{K} : |x - a|_1 < r^s\} = B_2(a, r) = \{x \in \mathbb{K} : |x - a|_2 < r\}$$

are the same. So the two topologies induced by $|\cdot|_1$ and $|\cdot|_2$ are the same.

Conversely, suppose the two topologies τ_1, τ_2 induced respectively by $|\cdot|_1$ and $|\cdot|_2$ are the same. First, we will show that $|x_n|_1 \to 0 \iff |x_n|_2 \to 0$ for a sequence $x_n \in \mathbb{K}$. By symmetry it is enough to show that $|x_n|_1 \to 0 \Rightarrow |x_n|_2 \to 0$. Let $\epsilon > 0$. Then $B_2(0, \epsilon)$ is open in $\tau_2 = \tau_1$, and $0 \in B_2(0, \epsilon)$, so there is $\epsilon' > 0$ such that

$$B_1(0, \epsilon') \subset B_2(0, \epsilon).$$

But $|x_n|_1 \to 0$, so there is N such that $x_n \in B_1(0, \epsilon') \subset B_2(0, \epsilon)$ for all $n > N$. So $|x_n|_2 \to 0$.

Second, $|x|_1 < 1 \iff |x|_2 < 1$. Indeed, $|x|_1 < 1$ iff $|x^n|_1 = |x|_1^n \to 0$ (if $|x|_1 \geq 1$ then $|x^n|_1 \geq 1$ for all n), iff $|x^n|_2 \to 0$ by the first step, iff $|x|_2 < 1$. Applying this to x^{-1}, we also get $|x|_1 > 1 \iff |x|_2 > 1$.

If $|\cdot|_1$ is trivial ($|x|_1 = 1$ for all $x \neq 0$), then by the second step so is $|\cdot|_2$, and they are equal. Otherwise, fix $y \in \mathbb{K}$ with $|y|_1 > 1$; then also $|y|_2 > 1$. Let $x \neq 0$ and write $|x|_1 = |y|_1^\alpha$ for some $\alpha \in \mathbb{R}$. For a rational number $\frac{m}{n} > \alpha$ (with $n > 0$) we have $|x|_1 < |y|_1^{m/n}$, so $\left|\frac{x^n}{y^m}\right|_1 < 1$, so $\left|\frac{x^n}{y^m}\right|_2 < 1$, so $|x|_2 < |y|_2^{m/n}$. In the same way, for $\frac{m}{n} < \alpha$ we get $|x|_2 > |y|_2^{m/n}$. Letting $\frac{m}{n} \to \alpha$ from both sides, $|x|_2 = |y|_2^\alpha$. Thus

$$\frac{\log|x|_1}{\log|y|_1} = \alpha = \frac{\log|x|_2}{\log|y|_2}, \quad \text{so} \quad \log|x|_1 = s \log|x|_2 \text{ with } s = \frac{\log|y|_1}{\log|y|_2} > 0,$$

i.e. $|x|_1 = |x|_2^s$ for all $x \in \mathbb{K}$. □

Proposition 2.6. *If $|\cdot|$ is a nontrivial valuation on $\mathbb{Q}$, then the valuation is either equivalent to the absolute value $\|\cdot\|$ or it is equivalent to $|\cdot|_p$ for some prime p.*

Proof. Let $|\cdot|$ be an archimedean valuation on $\mathbb{Q}$. I will use $\|\cdot\|$ to represent the absolute value. It is enough to show that $\frac{\log|n|}{\log n}$ is a fixed number for any $n \in \mathbb{N}$, $n \geq 2$. Let $m, n \in \mathbb{N}$ with $m, n \geq 2$. We can write n in base m: $n = a_0 + a_1 m + \cdots + a_r m^r$ with $0 \leq a_i < m$, where $r \leq \frac{\log n}{\log m}$. Since $|a_i| = |1 + \cdots + 1| \leq a_i < m$, we get

$$\begin{aligned} |n| &\leq \sum_{i=0}^{r} |a_i| \cdot |m|^i \\ &\leq \sum_{i=0}^{r} m \cdot \max\{1, |m|\}^r \\ &\leq m\,(r+1) \max\{1, |m|\}^{\frac{\log n}{\log m}}. \end{aligned}$$

If we replace n by n^k (so $r+1 \leq 1+k\frac{\log n}{\log m}$), we get

$$|n|^k \leq m\left(1+k\frac{\log n}{\log m}\right)\max\{1,|m|\}^{\frac{k\log n}{\log m}}$$
$$|n| \leq m^{\frac{1}{k}}\left(1+k\frac{\log n}{\log m}\right)^{\frac{1}{k}}\max\{1,|m|\}^{\frac{\log n}{\log m}}.$$

Letting $k\to\infty$ we get $|n| \leq \max\{1,|m|\}^{\frac{\log n}{\log m}}$. If $|m| \leq 1$ for some $m \geq 2$, this gives $|n| \leq 1$ for all n, so the valuation would be nonarchimedean. Hence $|m| > 1$ for all $m \geq 2$, and $|n| \leq |m|^{\frac{\log n}{\log m}}$. Then if we take the log of both sides we get $\frac{\log|n|}{\log n} \leq \frac{\log|m|}{\log m}$. Since m, n are arbitrary, we can swap them, so $\frac{\log|n|}{\log n}$ is a fixed number $s > 0$ for any $n \in \mathbb{N}$, $n \geq 2$, i.e. $|n| = n^s$. For $n \in \mathbb{Z}$ we will have $|n| = |-1|\,\|n\|^s = \|n\|^s$, and also $\left|\frac{1}{n}\right| = \frac{1}{|n|} = \left\|\frac{1}{n}\right\|^s$. Thus, for $\frac{a}{b} \in \mathbb{Q}$ we get

$$\left|\frac{a}{b}\right| = |a|\cdot\left|\frac{1}{b}\right| = \|a\|^s\cdot\left\|\frac{1}{b}\right\|^s = \left\|\frac{a}{b}\right\|^s.$$

Now suppose that the valuation is nonarchimedean, so $|n| \leq 1$ for all $n \in \mathbb{N}$ (Proposition 2.1). Let p be a prime such that $|p| < 1$. If such a prime does not exist, then every prime has $|q| = 1$, and by factorization $|r| = 1$ for all $r \in \mathbb{Q} - \{0\}$: the trivial valuation. So there exists a prime number p with $|p| < 1$. If we look at the set $\mathfrak{a} = \{x \in \mathbb{Z} : |x| < 1\}$, this is an ideal in the ring $\mathbb{Z}$ (by the strong triangle inequality) which contains the ideal $p\mathbb{Z}$; but the latter is a maximal ideal and $1 \notin \mathfrak{a}$, so $\mathfrak{a} = p\mathbb{Z}$. Thus for any prime $q \neq p$ we get $|q| = 1$. Let $|p| = t$ and $r \in \mathbb{Q}$, $r \neq 0$. We write $r = p^m a$ where p divides neither the numerator nor the denominator of a, so $|a| = 1$. Then

$$|r| = |p^m a| = |p|^m = t^m = (p^{-m})^{\frac{-\log t}{\log p}} = |r|_p^s,$$

where we set $s = \frac{-\log t}{\log p} > 0$. □

Example 2.7. Let $\mathbb{K}$ be a field. We can define a valuation on the field $\mathbb{K}(x)$ (the field of rational functions) in a similar way to $\mathbb{Q}_p$. Let $p(x)$ be an irreducible monic polynomial in $\mathbb{K}[x]$. To define the exponential valuation on $\mathbb{K}(x)$, let $f(x) \in \mathbb{K}(x)$, $f \neq 0$; then we can write

$$f(x) = p(x)^m\frac{g(x)}{h(x)}, \quad p(x)\nmid g(x)h(x).$$

Then $v_p(f) = m$. The multiplicative valuation would be defined as

$$|f(x)|_p = q^{-v_p(f)} = q^{-m}, \quad \text{where } q \in \mathbb{R} \text{ and } q > 1.$$

For $p(x) = x$, the completion of the field $\mathbb{K}(x)$ with respect to the nonarchimedean valuation $|\cdot|_x$ is the field $\mathbb{K}((x))$ of formal Laurent series, which contains the elements in the form

$$\mathbb{K}((x)) = \left\{\sum_{i=n}^{\infty} a_i x^i : n \in \mathbb{Z},\ a_i \in \mathbb{K}\right\}.$$

The field of formal Laurent series $\mathbb{K}((x))$ is the field of fractions of the ring $\mathbb{K}[[x]]$ of formal power series, which is

$$\mathbb{K}[[x]] = \left\{ \sum_{i=0}^{\infty} a_i x^i : a_i \in \mathbb{K} \right\}.$$

Using Hensel's lemma (*we will talk about it later*) we can prove that the algebraic closure of the field of formal Laurent series $\mathbb{K}((x))$, with coefficients from an algebraically closed field $\mathbb{K}$ of characteristic zero, is the field of Puiseux series $\mathbb{K}((x^*))$, which is the union

$$\bigcup_{n \in \mathbb{N}} \mathbb{K}((x^{\frac{1}{n}})).$$

Clearly $\mathbb{K}((x^{\frac{1}{n}})), \mathbb{K}((x^{\frac{1}{m}})) \subset \mathbb{K}((x^{\frac{1}{mn}}))$, so this union is a field. We define the *order* of an element $f(x) \in \mathbb{K}((x^*))$: if $f(x) = \sum_{i=n}^{\infty} a_i x^{\frac{i}{k}}$ with $a_n \neq 0$, then $\operatorname{ord}(f) = \frac{n}{k}$.

Proposition 2.8. *If $\mathbb{K}$ is a field with a nonarchimedean valuation $|\cdot|$ and the exponential valuation v (so $|x| = q^{-v(x)}$ for some $q > 1$), then*

- $\mathfrak{o} = \{x \in \mathbb{K} : |x| \leq 1\} = \{x \in \mathbb{K} : v(x) \geq 0\}$ *is a ring,*
- $\mathfrak{o}^* = \{x \in \mathbb{K} : |x| = 1\} = \{x \in \mathbb{K} : v(x) = 0\}$ *is the group of units of* $\mathfrak{o}$*,*
- $\mathfrak{p} = \{x \in \mathbb{K} : |x| < 1\} = \{x \in \mathbb{K} : v(x) > 0\}$ *is the unique maximal ideal of* $\mathfrak{o}$*.*

Proof. Let $x, y \in \mathfrak{o}$; then $|xy| = |x||y| \leq 1$ and $|x + y| \leq \max\{|x|, |y|\} \leq 1$. Now if $x \in \mathfrak{o}$, then since $|-1| = 1$ we get $|-x| \leq 1$. The other properties follow from the fact that $\mathbb{K}$ is a field, so $\mathfrak{o}$ is a ring. Let $x \in \mathfrak{o}^*$, so $|x| = 1$; then since $1 = |1| = |x \cdot x^{-1}| = |x| \cdot |x^{-1}| = |x^{-1}|$, we get that $x^{-1} \in \mathfrak{o}^*$. Conversely, if x and x^{-1} are both in $\mathfrak{o}$, then $|x| \leq 1$ and $\frac{1}{|x|} \leq 1$, so $|x| = 1$. So $\mathfrak{o}^*$ is exactly the group of units of $\mathfrak{o}$.

Now for the third part: if $x, y \in \mathfrak{p}$ then $|x + y| \leq \max\{|x|, |y|\} < 1$, and if $x \in \mathfrak{o}$ then for any $y \in \mathfrak{p}$ we get that $|xy| = |x||y| < 1$, since $|y| < 1$ and $|x| \leq 1$. So $\mathfrak{p}$ is an ideal. It is the unique maximal ideal: every element of $\mathfrak{o}$ which is not in $\mathfrak{p}$ has $|x| = 1$, so it is a unit; hence a proper ideal contains no element outside $\mathfrak{p}$, i.e. every proper ideal is contained in $\mathfrak{p}$. □

We say that a field $\mathbb{K}$ is *complete* with respect to a valuation $|\cdot|$ if every *Cauchy* sequence in $\mathbb{K}$ converges. An integral domain $\mathfrak{o}$ is called a *valuation ring* if for any $x \neq 0$ in $\mathbb{K}$, where $\mathbb{K}$ is its field of fractions, we have either $x \in \mathfrak{o}$ or $x^{-1} \in \mathfrak{o}$. An exponential valuation v is called a *discrete valuation* if there is $s > 0$ such that $v(\mathbb{K}^*) = s\mathbb{Z}$. A discrete valuation is called a *normalized valuation* if $s = 1$. For a normalized valuation, an element $\pi \in \mathfrak{o}$ is called a prime element if $v(\pi) = 1$, and for any $x \in \mathbb{K}^*$ we can write $x = u\pi^m$ where $m \in \mathbb{Z}$ and $u \in \mathfrak{o}^*$, i.e. $v(u) = 0$.

Example 2.9. The completion of the field $\mathbb{Q}$ with respect to the valuation $|\cdot|_p$ is the field of p-adic numbers. We can show that in a similar way to showing that the completion of $\mathbb{Q}$ with respect to the absolute value is $\mathbb{R}$.

Proposition 2.10. *If v is a discrete exponential valuation of $\mathbb{K}$, then*

$$\mathfrak{o} = \{x \in \mathbb{K} : v(x) \geq 0\}$$

is a principal ideal domain. Suppose v is normalized. Then the nonzero ideals of $\mathfrak{o}$ are given by $\mathfrak{p}^n = \pi^n\mathfrak{o} = \{x \in \mathbb{K} : v(x) \geq n\}$, $n \geq 0$, where π is a prime element, i.e. $v(\pi) = 1$, and one has

$$\mathfrak{p}^n/\mathfrak{p}^{n+1} \cong \mathfrak{o}/\mathfrak{p}.$$

Proof. The valuation is discrete, so there exists $s > 0$ such that $v(\mathbb{K}^*) = s\mathbb{Z}$. Let I be a nonzero ideal in $\mathfrak{o}$. Since the valuation is discrete and $v \geq 0$ on $\mathfrak{o}$, we can define

$$r = \min_{i \in I,\ i \neq 0} v(i) \in s\mathbb{Z}_{\geq 0},$$

and let $a \in I$ be an element such that $v(a) = r$. Let $x \in I$; then $v(x) \geq r$. Then the element $xa^{-1} \in \mathfrak{o}$, since $v(xa^{-1}) = v(x) + v(a^{-1}) = v(x) - r \geq 0$. Thus $x = (xa^{-1}) \cdot a$, which means $I = (a)$.

Suppose that v is a normalized discrete valuation of $\mathbb{K}$. Let I be a nonzero ideal of $\mathfrak{o}$. From the above it is a principal ideal, so there is $a \in I$ such that $I = (a)$. Now let $\pi \in \mathfrak{o}$ be such that $v(\pi) = 1$. Suppose $v(a) = m$. Then $v(\pi^{-m}a) = 0$, so $u = \pi^{-m}a \in \mathfrak{o}^*$. Thus $a = \pi^m \cdot u$, which means $I = (\pi^m) = \pi^m\mathfrak{o} = \{x : v(x) \geq m\}$.

To show the last result we define the homomorphism (of additive groups) $\phi : \mathfrak{p}^n \to \mathfrak{o}/\mathfrak{p}$ by $\pi^n x \mapsto x \pmod{\mathfrak{p}}$, $x \in \mathfrak{o}$. It is onto, and an element is in the kernel iff it is of the form $\pi^n \cdot (\pi u) = \pi^{n+1}u$ with $u \in \mathfrak{o}$. Hence the kernel is $\mathfrak{p}^{n+1}$, and that establishes the isomorphism $\mathfrak{p}^n/\mathfrak{p}^{n+1} \cong \mathfrak{o}/\mathfrak{p}$. □

Hensel's Lemma (second version). *Let $\mathbb{K}$ be a field which is complete with respect to a nonarchimedean valuation $|\cdot|$, with valuation ring $\mathfrak{o}$ and unique maximal ideal $\mathfrak{p}$. Suppose a primitive polynomial $f(x) \in \mathfrak{o}[x]$ (that is, $\max_i |a_i| = 1$ over its coefficients a_i) factors as*

$$f(x) \equiv g_0(x)\, h_0(x) \pmod{\mathfrak{p}},$$

where g_0, h_0 are relatively prime mod $\mathfrak{p}$. Then there exist $g(x)$ and $h(x)$ in $\mathfrak{o}[x]$ such that $f = gh$, $g \equiv g_0$, $h \equiv h_0 \pmod{\mathfrak{p}}$, $\deg(g) = \deg(g_0 \bmod \mathfrak{p})$ and $\deg(h) = \deg(f) - \deg(g)$.

Proof. We may choose g_0, h_0 so that $\deg(g_0) = m$ is the degree of $g_0 \bmod \mathfrak{p}$ (so the leading coefficient of g_0 is a unit) and, with $d = \deg(f)$, $\deg(h_0) \leq d - m$. Since g_0, h_0 are relatively prime mod $\mathfrak{p}$, there exist $a, b \in \mathfrak{o}[x]$ such that $g_0 a + h_0 b \equiv 1 \pmod{\mathfrak{p}}$. Now we look at the coefficients of the two polynomials

$$f - g_0 h_0, \qquad g_0 a + h_0 b - 1.$$

Then we pick the coefficient with the least valuation v (the largest $|\cdot|$) and call it π; it is in $\mathfrak{p}$. (If all these coefficients are 0, then $f = g_0 h_0$ and we are done.) We want to find, by induction,

$g_1, g_2, \cdots \in \mathfrak{o}[x]$ with $\deg(g_i) < m$, and $h_1, h_2, \cdots \in \mathfrak{o}[x]$ with $\deg(h_i) \leq d - m$, such that

$$g = g_0 + g_1\pi + g_2\pi^2 + \cdots$$
$$h = h_0 + h_1\pi + h_2\pi^2 + \cdots$$

and $f \equiv G_n H_n \pmod{\pi^n}$, where

$$G_n = g_0 + g_1\pi + \cdots + g_{n-1}\pi^{n-1},$$
$$H_n = h_0 + h_1\pi + \cdots + h_{n-1}\pi^{n-1}.$$

(Here $\equiv \pmod{\pi^n}$ means that the difference has all its coefficients in $\pi^n\mathfrak{o}$.)

For $n = 1$ we have $f \equiv G_1 H_1 = g_0 h_0 \pmod{\pi}$ by the choice of π. Suppose $f \equiv G_n H_n \pmod{\pi^n}$; we want to find $g_n, h_n \in \mathfrak{o}[x]$ such that

$$f \equiv (G_n + g_n\pi^n)(H_n + h_n\pi^n) \pmod{\pi^{n+1}}.$$

Expanding, the terms with π exponent more than or equal to $n+1$ will be zero, and $G_n \equiv g_0$, $H_n \equiv h_0 \pmod{\pi}$:

$$\begin{aligned} f &\equiv G_n H_n + g_0 h_n \pi^n + g_n h_0 \pi^n && \pmod{\pi^{n+1}} \\ f - G_n H_n &\equiv \pi^n (g_0 h_n + g_n h_0) && \pmod{\pi^{n+1}} \\ \frac{f - G_n H_n}{\pi^n} &\equiv g_0 h_n + g_n h_0 && \pmod{\pi}. \end{aligned}$$

But we also have $g_0 a + h_0 b \equiv 1 \pmod{\pi}$; multiplying by $L(x) = \frac{f - G_n H_n}{\pi^n} \in \mathfrak{o}[x]$ we get

$$L a g_0 + L b h_0 \equiv L \pmod{\pi}.$$

Use Euclid division (possible because the leading coefficient of g_0 is a unit): $Lb = q(x) g_0(x) + r(x)$ with $\deg(r) < \deg(g_0) = m$. We let $g_n(x) = r(x)$; then we get

$$L a g_0 + h_0(q g_0 + g_n) = g_0(La + q h_0) + h_0 g_n \equiv L \pmod{\pi}.$$

Let h_n be $La + q h_0$ with its coefficients in $\pi\mathfrak{o}$ removed. Then $g_0 h_n + h_0 g_n \equiv L \pmod{\pi}$. Moreover $\deg(L) \leq d$ and $\deg(h_0 g_n) < (d - m) + m = d$, so $g_0 h_n \equiv L - h_0 g_n$ has degree at most d mod π; since the leading coefficient of g_0 is a unit, $\deg(h_n) \leq d - m$.

Since $\mathbb{K}$ is complete and $|\pi^n| \to 0$, the coefficients of G_n and H_n converge, to polynomials g and h with $\deg(g) = m$ and $\deg(h) \leq d - m$, and $f \equiv gh \pmod{\pi^n}$ for every n, so $f = gh$. Comparing degrees, $\deg(h) = d - m$. □

Proposition 2.11. *Let $\mathbb{K}$ be an algebraically closed field with characteristic zero. Then the algebraic closure of the field $\mathbb{K}((x))$ is the Puiseux series field $\mathbb{K}((x^*)) = \bigcup_n \mathbb{K}((x^{\frac{1}{n}}))$. The main ideas of the proof are from [3].*

Proof. The field $\mathbb{K}((x))$ is not algebraically closed, because of $F(x,Y) = Y^2 - x$: clearly $x \in \mathbb{K}((x))$ but $\pm x^{\frac{1}{2}} \notin \mathbb{K}((x))$. Every element of $\mathbb{K}((x^{\frac{1}{k}}))$ is algebraic over $\mathbb{K}((x))$, since $\mathbb{K}((x^{\frac{1}{k}})) = \mathbb{K}((x))(x^{\frac{1}{k}})$ is an extension of degree k. Thus if we show that $\mathbb{K}((x^*))$ is algebraically closed, we are done.

It is enough to show that any monic polynomial of degree $n \geq 2$ in $\mathbb{K}((x^*))[Y]$ factors into two polynomials of smaller degree; then, by induction on the degree, every polynomial factors into linear factors. Let

$$F(x,Y) = Y^n + \left(\sum_{i=r_1}^{\infty} a_{1i} x^{\frac{i}{k_1}}\right) Y^{n-1} + \cdots + \left(\sum_{i=r_n}^{\infty} a_{ni} x^{\frac{i}{k_n}}\right), \quad a_{ji} \in \mathbb{K}.$$

Using the Tschirnhausen transformation $Y = Y' - \frac{\text{coefficient of } Y^{n-1}}{n}$ (possible since the characteristic is zero) we can assume that the coefficient of Y^{n-1} is zero. We can also assume that the constant term is not zero (otherwise Y is a factor). Thus we have

$$F(x,Y) = Y^n + \left(\sum_{i=r_2}^{\infty} a_{2i} x^{\frac{i}{k_2}}\right) Y^{n-2} + \cdots + \left(\sum_{i=r_n}^{\infty} a_{ni} x^{\frac{i}{k_n}}\right),$$

where, for each coefficient c_j which is not zero, $a_{jr_j} \neq 0$, so $\operatorname{ord}(c_j) = \frac{r_j}{k_j}$. Let $r = \min_j \frac{r_j}{jk_j}$ over the nonzero coefficients c_j. Using a suitable substitution $z^m = x$, for example $m = \operatorname{LCM}_j(jk_j)$, let $m_j = \frac{m}{k_j}$; then mr is an integer and

$$F(z,Y) = Y^n + \left(\sum_{i=r_2}^{\infty} a_{2i} z^{im_2}\right) Y^{n-2} + \cdots + \left(\sum_{i=r_n}^{\infty} a_{ni} z^{im_n}\right).$$

Set $Y = z^{mr}U$. Then we get

$$\begin{aligned} F(z, z^{mr}U) &= U^n z^{nmr} + \left(\sum_{i=r_2}^{\infty} a_{2i} z^{im_2}\right) z^{(n-2)mr} U^{n-2} + \cdots + \left(\sum_{i=r_n}^{\infty} a_{ni} z^{im_n}\right) \\ &= z^{nmr}\left[U^n + \left(\sum_{i=r_2}^{\infty} a_{2i} z^{im_2 - 2mr}\right) U^{n-2} + \cdots + \left(\sum_{i=r_n}^{\infty} a_{ni} z^{im_n - nmr}\right)\right]. \end{aligned}$$

The exponent of z in each coefficient is non-negative: the smallest exponent is $r_j m_j - jmr \geq r_j \frac{m}{k_j} - jm \frac{r_j}{jk_j} = 0$, with equality for the j where the minimum r is attained. Now consider the polynomial

$$G(z,U) = U^n + \left(\sum_{i=r_2}^{\infty} a_{2i} z^{im_2 - 2mr}\right) U^{n-2} + \cdots + \left(\sum_{i=r_n}^{\infty} a_{ni} z^{im_n - nmr}\right) \in \mathbb{K}[[z]][U].$$

If we look at $G(0,U)$, this polynomial factors into two relatively prime polynomials, because $\mathbb{K}$ is algebraically closed and the coefficient of U^{n-1} is zero. Indeed, $G(0,U)$ is not U^n, since the coefficient where the minimum r is attained has a nonzero constant term. If $G(0,U)$ had only one root α, then $G(0,U) = (U-\alpha)^n$, whose coefficient of U^{n-1} is $-n\alpha = 0$, so $\alpha = 0$ (characteristic zero) and $G(0,U) = U^n$, a contradiction. So $G(0,U)$ has two different roots

$\alpha \neq \beta$, and $G(0, U) = (U - \alpha)^e h_0(U)$ with $h_0(\alpha) \neq 0$ and $1 \leq e < n$: two relatively prime factors. Hence, using Hensel's lemma (second version) for the complete field $\mathbb{K}((z))$, with valuation ring $\mathbb{K}[[z]]$ and $\mathbb{K}[[z]]/(z) = \mathbb{K}$, we get $G = gh$ with $\deg(g) = e$ and $\deg(h) = n - e$. Then $F(z, Y) = z^{nmr} G(z, z^{-mr} Y)$ factors into two polynomials of degrees e and $n - e$ over $\mathbb{K}((z)) = \mathbb{K}((x^{\frac{1}{m}})) \subset \mathbb{K}((x^*))$, which is the desired. □

Ostrowski's Theorem. *Let $\mathbb{K}$ be a field of characteristic 0. Suppose that it is complete with respect to an archimedean valuation $|\cdot|$. Then $\mathbb{K}$ is isomorphic to either $\mathbb{R}$ or $\mathbb{C}$.*

Proof. First, I will show that $\mathbb{K}$ contains a subfield isomorphic to $\mathbb{R}$. Let $1_{\mathbb{K}}$ be the multiplicative identity of $\mathbb{K}$; then we consider the ring homomorphism

$$\phi : \mathbb{Z} \to \mathbb{K}, \quad \phi(1) = 1_{\mathbb{K}},$$

and $\phi(n) = n \cdot 1_{\mathbb{K}}$. It is one to one because the characteristic is 0. Thus $\mathbb{K}$ contains a copy of $\mathbb{Z}$. Now since $\underbrace{1_{\mathbb{K}} + 1_{\mathbb{K}} + \cdots + 1_{\mathbb{K}}}_{n \text{ times}} = n1_{\mathbb{K}} \in \mathbb{K}$ is not zero, its multiplicative inverse is contained in $\mathbb{K}$, and it can be represented as $\frac{1}{n}1_{\mathbb{K}}$. The homomorphism ϕ can be extended to $\mathbb{Q}$: we map $\frac{1}{n}$ to $\frac{1}{n} \cdot 1_{\mathbb{K}}$, so by that $\mathbb{K}$ contains a copy of $\mathbb{Q}$, and

$$\phi : \mathbb{Q} \to \mathbb{K}$$

will be a homomorphism between fields. Now to show that $\mathbb{K}$ contains a copy of $\mathbb{R}$, we define a valuation on $\mathbb{Q}$, call it $|\cdot|_{\mathbb{Q}}$, as follows:

$$|\cdot|_{\mathbb{Q}} : \mathbb{Q} \to \mathbb{R}, \quad |r|_{\mathbb{Q}} = |\phi(r)|_{\mathbb{K}}.$$

This defines an archimedean valuation on $\mathbb{Q}$ (the values $|n \cdot 1_{\mathbb{K}}|$ are not bounded). But we proved earlier (Proposition 2.6) that any archimedean valuation on $\mathbb{Q}$ is equivalent to $\|\cdot\|$, the absolute value; thus $|\cdot|_{\mathbb{Q}} = \|\cdot\|^s$ for some $s \in \mathbb{R}^+$. Now let $v \in \mathbb{R}$; then there exists a sequence of rational numbers $r_n \to v$ with respect to the absolute value. Then $\phi(r_n)$ is a Cauchy sequence in $\mathbb{K}$, since $|\phi(r_n) - \phi(r_m)|_{\mathbb{K}} = |r_n - r_m|_{\mathbb{Q}} = \|r_n - r_m\|^s$, so it converges, since $\mathbb{K}$ is complete, and we set $\phi(v) = \lim \phi(r_n)$ (the limit does not depend on the sequence). Hence $\mathbb{K}$ contains a copy of $\mathbb{R}$, on which $|\cdot|_{\mathbb{K}} = \|\cdot\|^s$. Without loss of generality we can assume $\mathbb{R} \subset \mathbb{K}$, with $|x| = \|x\|^s$ for $x \in \mathbb{R}$.

Now we will show that any $\xi \in \mathbb{K}$ satisfies a quadratic polynomial in $\mathbb{R}[x]$. Consider the following function $f : \mathbb{C} \to \mathbb{R}$:

$$f(z) = |\xi^2 - (z + \bar{z})\xi + z\bar{z}|_{\mathbb{K}}.$$

Note that $z + \bar{z}$ and $z\bar{z}$ are real numbers, so the expression is in $\mathbb{K}$. The function $f(z)$ is continuous: the maps $z \mapsto z + \bar{z}$ and $z \mapsto z\bar{z}$ are continuous from $\mathbb{C}$ to $\mathbb{R}$, and f is obtained from them by the operations of $\mathbb{K}$ and the valuation, which are continuous (for example $\big|\, |a|_{\mathbb{K}} - |b|_{\mathbb{K}} \,\big| \leq |a - b|_{\mathbb{K}}$). Also $f(z) \geq \|z\|^{2s} - \|z + \bar{z}\|^s \, |\xi| - |\xi|^2 \to \infty$ as $\|z\| \to \infty$. Hence $f(z)$ has a minimum m; define

$$S = \{z \in \mathbb{C} : f(z) = m\}.$$

If we prove that $m = 0$, then we will have $z \in \mathbb{C}$ such that $f(z) = 0$, i.e. $\xi^2 - (z + \overline{z})\xi + z\overline{z} = 0$. Suppose, on the contrary, that $m > 0$. The set S is nonempty, bounded and closed. It is bounded because $\lim_{\|z\| \to \infty} f(z) = \infty$, and it is closed because it is the inverse image of a closed set. Now if we look at $\{\|z\| : z \in S\}$, this is a closed bounded subset of $\mathbb{R}$, so it has a maximum, attained at some $z_0 \in S$. Let $\epsilon \in \mathbb{R}$ be such that $\epsilon > 0$ and $|\epsilon| = \epsilon^s < m$, and define the function

$$h(x) = x^2 - (z_0 + \overline{z}_0)x + z_0\overline{z}_0 + \epsilon.$$

The function $h(x)$ has no real roots, because it has a negative discriminant: $(z_0 + \overline{z}_0)^2 - 4(z_0\overline{z}_0 + \epsilon) = (z_0 - \overline{z}_0)^2 - 4\epsilon = -4(\operatorname{Im} z_0)^2 - 4\epsilon < 0$. Let $z_1, \overline{z}_1 \in \mathbb{C}$ be the two complex roots of $h(x)$. Since the constant term of a quadratic polynomial is the product of its roots, we get that $z_1\overline{z}_1 = z_0\overline{z}_0 + \epsilon$. Thus $\|z_1\| > \|z_0\|$, so $z_1 \notin S$, i.e. $f(z_1) > m$.

Define the function

$$G(x) = (h(x) - \epsilon)^n - (-\epsilon)^n.$$

Then $G(z_1) = (h(z_1) - \epsilon)^n - (-\epsilon)^n = (-\epsilon)^n - (-\epsilon)^n = 0$. $G(x)$ is a real polynomial of degree $2n$; suppose $G(x)$ has the roots $\alpha_1, \cdots, \alpha_{2n}$ and assume $z_1 = \alpha_1$. Then $G(x)$ will factor completely as

$$G(x) = \prod_{i=1}^{2n} (x - \alpha_i).$$

But since G has real coefficients, we also have $G(x) = \prod(x - \overline{\alpha}_i)$. Thus

$$(G(x))^2 = \prod(x - \alpha_i) \prod(x - \overline{\alpha}_i) = \prod \left(x^2 - (\alpha_i + \overline{\alpha}_i)x + \alpha_i\overline{\alpha}_i\right).$$

Since $f(z) = |\xi^2 - (z + \overline{z})\xi + z\overline{z}| \geq m$ for any $z \in \mathbb{C}$, we get that

$$|G(\xi)|^2 = \prod_{i=1}^{2n} f(\alpha_i) \geq f(z_1)\, m^{2n-1}.$$

Also we have, since $h(\xi) - \epsilon = \xi^2 - (z_0 + \overline{z}_0)\xi + z_0\overline{z}_0$,

$$|G(\xi)| = |(h(\xi) - \epsilon)^n - (-\epsilon)^n| \leq |h(\xi) - \epsilon|^n + |\epsilon|^n = (f(z_0))^n + |\epsilon|^n = m^n + |\epsilon|^n.$$

Combining the two inequalities we get

$$\begin{aligned} f(z_1)\, m^{2n-1} &\leq (m^n + |\epsilon|^n)^2 \\ \frac{f(z_1)}{m} &\leq \left(1 + \left(\frac{|\epsilon|}{m}\right)^n\right)^2. \end{aligned}$$

Note that $0 < |\epsilon| < m$, so $\frac{|\epsilon|}{m} < 1$. Thus if we take the limit as $n \to \infty$ we get

$$f(z_1) \leq m.$$

But then $z_1 \in S$, which contradicts the fact that $\|z_1\| > \|z_0\|$. So $m = 0$.

Finally, every $\xi \in \mathbb{K}$ is a root of a real polynomial of degree 2. If $\mathbb{K} \neq \mathbb{R}$, take $\xi \in \mathbb{K} - \mathbb{R}$; then $\mathbb{R}(\xi)$ is an extension of degree 2 of $\mathbb{R}$, so $\mathbb{R}(\xi) \cong \mathbb{C}$, and $\mathbb{K}$ contains a copy of $\mathbb{C}$. Every element of $\mathbb{K}$ is algebraic over this copy of $\mathbb{C}$, which is algebraically closed, so $\mathbb{K} \cong \mathbb{C}$. Otherwise $\mathbb{K} = \mathbb{R}$. □

3 Extension of valuations

If we take a field $\mathbb{K}$ which is complete with respect to a valuation $|\cdot|$, archimedean or nonarchimedean, and consider an algebraic extension $\mathbb{L}|\mathbb{K}$, we can extend the valuation $|\cdot|$ to $\mathbb{L}$ uniquely. In the following example we will see the case of a quadratic extension.

Example 3.1. Let $\mathbb{K}$ be a complete field with valuation $|\cdot|_{\mathbb{K}}$, of characteristic not 2. Suppose $p(x) = x^2 + bx + c$ is an irreducible polynomial in $\mathbb{K}[x]$. Let $\theta = \sqrt{b^2 - 4c}$; then the polynomial factors completely in $\mathbb{K}(\theta)[x]$ as

$$x^2 + bx + c = \left(x - \frac{-b-\theta}{2}\right)\left(x - \frac{-b+\theta}{2}\right).$$

Let $\mathbb{L} = \mathbb{K}(\theta) = \{\alpha + \theta\beta : \alpha, \beta \in \mathbb{K}\}$. We want to extend the valuation $|\cdot|_{\mathbb{K}}$ to the extension $\mathbb{L}$ as $|\cdot|_{\mathbb{L}}$, and we want the extension of the valuation to restrict to the field $\mathbb{K}$, i.e. for any $\gamma \in \mathbb{K}$, $|\gamma|_{\mathbb{K}} = |\gamma|_{\mathbb{L}}$. If $\alpha + \theta\beta \in \mathbb{L}$, then $(\alpha + \theta\beta)(\alpha - \theta\beta) = \alpha^2 - (b^2 - 4c)\beta^2 \in \mathbb{K}$, so we have

$$|(\alpha + \theta\beta)(\alpha - \theta\beta)|_{\mathbb{K}} = |\alpha + \theta\beta|_{\mathbb{L}}\,|\alpha - \theta\beta|_{\mathbb{L}}. \tag{3}$$

If $\alpha = 0$, $\beta = 1$, we get $|\theta|_{\mathbb{L}}\,|-\theta|_{\mathbb{L}} = |\theta|_{\mathbb{L}}^2 = |-(b^2-4c)|_{\mathbb{K}}$, so $|\theta|_{\mathbb{L}} = \sqrt{|b^2-4c|_{\mathbb{K}}}$. Note that $|\theta|_{\mathbb{L}}^2 = |b^2 - 4c|_{\mathbb{K}}$ is the valuation of the constant term in the minimal polynomial of θ, which is $x^2 - (b^2 - 4c)$.

Definition. Let $\mathbb{L}|\mathbb{K}$ be an algebraic extension. If $\alpha \in \mathbb{L}$, then it is algebraic over $\mathbb{K}$. Suppose $[\mathbb{K}(\alpha) : \mathbb{K}] = n$, with minimal polynomial

$$p(x) = x^n + c_1 x^{n-1} + \cdots + c_n, \quad \text{and } p(\alpha) = 0.$$

We define the *norm* of α as $N_{\mathbb{K}(\alpha)|\mathbb{K}}(\alpha) = \det(A_\alpha)$, where A_α is the linear transformation $A_\alpha : \mathbb{K}(\alpha) \to \mathbb{K}(\alpha)$, $A_\alpha(x) = \alpha x$, considering $\mathbb{K}(\alpha)$ as a vector space over $\mathbb{K}$ with basis $\{1, \alpha, \alpha^2, \cdots, \alpha^{n-1}\}$. Let $\alpha^i = (0, \cdots, 1, 0, \cdots, 0)^T$ be the vector with 1 in the $(i+1)$-th coordinate and the remaining zeroes. Then A_α will send $\alpha^i \mapsto \alpha^{i+1}$, except $\alpha^{n-1} \mapsto \alpha^n = -c_n - \cdots - c_1\alpha^{n-1}$. Thus

$$A_\alpha = \begin{pmatrix} 0 & 0 & \cdots & 0 & -c_n \\ 1 & 0 & \cdots & 0 & -c_{n-1} \\ 0 & 1 & \cdots & 0 & -c_{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_1 \end{pmatrix}, \qquad N_{\mathbb{K}(\alpha)|\mathbb{K}}(\alpha) = \det(A_\alpha) = (-1)^n c_n.$$

Theorem 3.2. *Let $\mathbb{K}$ be a complete field with respect to the valuation $|\cdot|$. Suppose $\mathbb{L}|\mathbb{K}$ is an algebraic extension of $\mathbb{K}$. Then the valuation $|\cdot|$ extends uniquely to $\mathbb{L}$, by the formula*

$$|\alpha|_{\mathbb{L}} = \sqrt[n]{|N_{\mathbb{K}(\alpha)|\mathbb{K}}(\alpha)|_{\mathbb{K}}},$$

where $n = [\mathbb{K}(\alpha) : \mathbb{K}]$.

Proof. Suppose the valuation $|\cdot|_{\mathbb{K}}$ is nonarchimedean, and suppose $\mathbb{L}|\mathbb{K}$ is an algebraic extension. Let $\mathfrak{o}$ be the valuation ring of $\mathbb{K}$. First, we need two facts.

(a) *If* $g(x) = a_d x^d + \cdots + a_0 \in \mathbb{K}[x]$ *is irreducible with* $a_0 a_d \neq 0$, *then* $\max_i |a_i| = \max\{|a_0|, |a_d|\}$. Multiplying by a constant, we can assume $\max_i |a_i| = 1$, i.e. g is primitive. If $\max\{|a_0|, |a_d|\} < 1$, let r be the smallest index with $|a_r| = 1$; then $0 < r < d$ and

$$g(x) \equiv x^r(a_r + a_{r+1}x + \cdots + a_d x^{d-r}) \pmod{\mathfrak{p}},$$

two relatively prime factors (the second has nonzero constant term $a_r \bmod \mathfrak{p}$). By Hensel's lemma (second version) g is a product of polynomials of degrees r and $d - r$, which contradicts that g is irreducible.

(b) *For* $\alpha \in \mathbb{L}$: $|N_{\mathbb{K}(\alpha)|\mathbb{K}}(\alpha)| \leq 1$ *if and only if the minimal polynomial of* α *has all its coefficients in* $\mathfrak{o}$. Indeed, by (a) applied to the minimal polynomial $p(x)$ (with $a_d = 1$), its coefficients are all in $\mathfrak{o}$ iff $|c_n| \leq 1$, and $|c_n| = |N_{\mathbb{K}(\alpha)|\mathbb{K}}(\alpha)|$.

Now we show that $|\alpha|_{\mathbb{L}} = |N_{\mathbb{K}(\alpha)|\mathbb{K}}(\alpha)|^{1/n}$ defines a nonarchimedean valuation on $\mathbb{L}$ which extends $|\cdot|$. Suppose $\alpha, \beta \in \mathbb{L}$.

- If $\alpha \in \mathbb{K}$ then $n = 1$ and $N(\alpha) = \alpha$, so $|\alpha|_{\mathbb{L}} = |\alpha|$: it extends $|\cdot|$.
- $|\alpha|_{\mathbb{L}} = 0$ iff $N(\alpha) = 0$ iff $c_n = 0$ iff $\alpha = 0$ (the minimal polynomial is irreducible).
- $|\alpha\beta|_{\mathbb{L}} = |\alpha|_{\mathbb{L}}|\beta|_{\mathbb{L}}$: let $\mathbb{M} = \mathbb{K}(\alpha, \beta)$, a finite extension. For $\gamma \in \mathbb{M}$, the matrix of multiplication by γ on $\mathbb{M}$ is made of $[\mathbb{M} : \mathbb{K}(\gamma)]$ copies of A_γ (in a basis $\{\mu_j \gamma^i\}$ with $\{\mu_j\}$ a basis of $\mathbb{M}$ over $\mathbb{K}(\gamma)$), so $N_{\mathbb{M}|\mathbb{K}}(\gamma) = N_{\mathbb{K}(\gamma)|\mathbb{K}}(\gamma)^{[\mathbb{M}:\mathbb{K}(\gamma)]}$ and

  $$|\gamma|_{\mathbb{L}} = |N_{\mathbb{M}|\mathbb{K}}(\gamma)|^{1/[\mathbb{M}:\mathbb{K}]}.$$

  Since $N_{\mathbb{M}|\mathbb{K}}(\alpha\beta) = \det(A_{\alpha\beta}) = \det(A_\alpha A_\beta) = N_{\mathbb{M}|\mathbb{K}}(\alpha)\, N_{\mathbb{M}|\mathbb{K}}(\beta)$ (multiplication maps on $\mathbb{M}$), the valuation is multiplicative.
- $|\alpha + \beta|_{\mathbb{L}} \leq \max\{|\alpha|_{\mathbb{L}}, |\beta|_{\mathbb{L}}\}$: we can suppose $|\alpha|_{\mathbb{L}} \leq |\beta|_{\mathbb{L}}$, $\beta \neq 0$, and divide by β; so we have to show that $|\gamma|_{\mathbb{L}} \leq 1 \Rightarrow |\gamma + 1|_{\mathbb{L}} \leq 1$. If $|\gamma|_{\mathbb{L}} \leq 1$, then by (b) the minimal polynomial $p(x)$ of γ has its coefficients in $\mathfrak{o}$. The minimal polynomial of $\gamma + 1$ is $p(x - 1)$, which also has its coefficients in $\mathfrak{o}$, so by (b) again $|\gamma + 1|_{\mathbb{L}} \leq 1$.

Uniqueness: suppose $|\cdot|'$ is another valuation on $\mathbb{L}$ extending $|\cdot|$. First, if $|\alpha|_{\mathbb{L}} \leq 1$ then $|\alpha|' \leq 1$. Indeed, α satisfies $\alpha^n + c_1\alpha^{n-1} + \cdots + c_n = 0$ with $c_i \in \mathfrak{o}$ by (b). If $|\alpha|' > 1$, then dividing by α^n,

$$1 = -c_1\alpha^{-1} - \cdots - c_n\alpha^{-n},$$

and the right side has $|\cdot|' < 1$, since $|c_i|' = |c_i| \leq 1$ and $|\alpha^{-1}|' < 1$ (we use that $|\cdot|'$ is nonarchimedean, which holds because it is bounded on $\mathbb{N} \subset \mathbb{K}$ by Proposition 2.1). This is a contradiction. Now let $\alpha \neq 0$, $n = [\mathbb{K}(\alpha) : \mathbb{K}]$ and $a = N_{\mathbb{K}(\alpha)|\mathbb{K}}(\alpha) \in \mathbb{K}$. Then $|\alpha^n/a|_{\mathbb{L}} = |\alpha|_{\mathbb{L}}^n/|a| = 1$, so $|\alpha^n/a|' \leq 1$, i.e. $(|\alpha|')^n \leq |a|' = |a| = |\alpha|_{\mathbb{L}}^n$, so $|\alpha|' \leq |\alpha|_{\mathbb{L}}$. Doing the same for α^{-1} gives $|\alpha|' \geq |\alpha|_{\mathbb{L}}$. So $|\cdot|' = |\cdot|_{\mathbb{L}}$.

Finally, if $|\cdot|$ is archimedean, then by Ostrowski's theorem $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, with $|x| = \|x\|^s$ on $\mathbb{R}$ (as in its proof). Here $s \leq 1$, since $2^s = |1+1| \leq |1| + |1| = 2$. If $\mathbb{K} = \mathbb{C}$ there is nothing to prove, because $\mathbb{C}$ is algebraically closed, so $\mathbb{L} = \mathbb{C}$. If $\mathbb{K} = \mathbb{R}$, the only algebraic extension which is not trivial is $\mathbb{C}|\mathbb{R}$. There, for $z \notin \mathbb{R}$, the minimal polynomial of z is $x^2 - (z+\bar{z})x + z\bar{z}$, so $N_{\mathbb{C}|\mathbb{R}}(z) = z\bar{z}$, and the formula gives $|z|_{\mathbb{C}} = \sqrt{\|z\bar{z}\|^s} = \|z\|^s$ (also for $z \in \mathbb{R}$). This is a valuation: the only property which is not clear is the triangle inequality, and it holds because $\|z+w\|^s \leq (\|z\| + \|w\|)^s \leq \|z\|^s + \|w\|^s$ for $0 < s \leq 1$.

For uniqueness, let $|\cdot|'$ be any valuation on $\mathbb{C}$ extending $|\cdot|$. Since $(|i|')^2 = |-1| = 1$, we have $|i|' = 1$. For $z = a + bi$ with $a, b \in \mathbb{R}$,

$$|z|' \leq |a| + |b|\,|i|' = \|a\|^s + \|b\|^s \leq 2\|z\|^s = 2|z|_{\mathbb{C}}.$$

Applying this to z^k gives $(|z|')^k \leq 2|z|_{\mathbb{C}}^k$, i.e. $|z|' \leq 2^{\frac{1}{k}}|z|_{\mathbb{C}}$, and letting $k \to \infty$, $|z|' \leq |z|_{\mathbb{C}}$. Applying the same to z^{-1} (for $z \neq 0$) gives $|z|' \geq |z|_{\mathbb{C}}$. So $|\cdot|' = |\cdot|_{\mathbb{C}}$. □

References

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Appendix: the p-adics Python codes

The first code converts rational numbers into p-adics. The second code checks if a number has a square root in the p-adics.

The first code (the file `padic_tool.py`):

```
if __name__ == '__main__':
    print("This tool to evaluate the first ‘n‘ coefficients of the p-adic for a rational number
    ‘a/b‘")

# Enter the numerator of the fraction
    loop = True
    while loop:
        try:
            a = int(input("Enter the numerator of the fraction 'a':"))
            loop = False
        except ValueError:
            print("Please enter an integer value!")

# Enter the denominator of the fraction
    loop = True
    while loop:
        try:
```

```
            b = int(input("Enter the denominator of the fraction 'b':"))
            if b == 0:
                print("The denominator cannot be 0!")
            else:
                loop = False
        except ValueError:
            print("Please enter an integer value!")
    d = b

# Enter the prime p for the p-adic
    loop = True
    while loop:
        try:
            p = int(input("Enter the 'p' value:"))
            if p < 2 or any(p % q == 0 for q in range(2, int(p**0.5) + 1)):
                print("'p' must be a prime number!")
            else:
                loop = False
        except ValueError:
            print("PLease enter an integer value!")

# Enter the number of coefficients
    loop = True
    while loop:
        try:
            n = int(input("Enter a positive 'n' for the number of coefficients you want to
    evaluate:"))
            if n < 1:
                print("Please enter a positive integer! ")
            else:
                loop = False
        except ValueError:
            print("Please enter a positive integer! ")
    c = int(a)
    if a == 0:
        print("0 is 0 in every p-adic: all its coefficients are 0.")
        raise SystemExit

# The following code is to find the equivalent simplified fraction of a/b as r/s
    r = a
    s = b
    if r < 0:
        r = -r
    if s < 0:
        s = -s
    if r >= s:
        while r % s !=0:
            A = r
            B = s
            while A % B !=0:
                r = B
                s = A % B
                break
        g = s
    if s >= r:
```

```
        while s % r != 0:
            A = r
            B = s
            while B % A != 0:
                s = A
                r = B % A
                break
        g = r
#-----------------------------------------
# The following two while's are to make count the p powers in a and b then clear them of it
    k = 0
    while a % p**(k+1) == 0:
        k = k + 1
    a = a // p**k

    l = 0
    while b % p**(l+1) == 0:
        l = l + 1
    b = b // p**l
# The following variable "m" is max power of p that divides a/b
    m = k - l

# The result stand for the evaluated p-adic and j for the counter for the number of coefficients
    result = ""
    j = 0


    while j < n:
        i = 0
        while i < p:
            if (a - i * b) % p == 0:
                a = (a - i * b) // p
                result = result + " " + str(int(i))
                break
            i = i + 1
        j = j + 1

#msg = str(‘the first‘, int(n),‘,‘, int(p), ‘-adic coefficients of‘ )
#msg += str(int(c), ‘/‘, int(d), "=", int(r), ‘/‘, int(s), ‘are‘)
    print(".")
    print("The fraction", int(c), "/", int(d), " is equivalent to ", c // g, "/", d // g)

    print("The first", int(n), "coefficients of ", int(p), "-adic for", int(c), "/", int(d), "
    are")
    print(str(result))
    print("NOTE!: The first entry is the coefficient of p to the power", int(m))
```

The second code (the file `quadratic.py`):

```
print("This tool is to check if an integer have a square root in p-adic")
p = int(input("Enter the prime for the p-adic:"))
while p < 3 or any(p % q == 0 for q in range(2, int(p**0.5) + 1)):
    p = int(input("p must be an odd prime (for p = 2 this method does not work). Enter p:"))
```

```
i = 0
while i < 2:
    a = int(input("Enter a number ‘a‘ less than the prime you entered to check it have a square
    root :"))
    if 0 <= a < p:
        break
s = []
i = 0
while i < p:
    s = s + [i]
    i = i + 1
t = []
i = 0
while i < p:
    t = t + [int(s[i])**2 % p]
    i = i + 1


i = 0
while i < 2:
    if a in t:
        break
    print("The ‘a‘ you entered is not a square in ‘p‘-adic")
    a = int(input("Enter another ‘a‘:"))


print("The number you entered", int(a), "have a square root in the", int(p),"-adic")
n = int(input("How many coefficients you want to evaluate?"))

i = 0
while i < p:
    if a == t[i]:
        x0 = i
        break
    i = i+1
y = [x0]
z = x0
j=0
while j<n-1:
    k = 0
    while k<p:
        T = (z**2-a)//p**(j+1)
        S = T + 2*z*k
        if S % p == 0:
            y = y +[k]
            z = z +k * p**(j+1)
            break
        k = k + 1
    j = j + 1
print("The square root of", int(a), "in the ", int(p), "-adic is:")
print(y)
print("The first number is the coefficient of p^0, the second for p^1 and so on...")
print("NOTE: if the ‘a‘ you entered is not a square in the integers then its p-adic is infinite!
    ")
End = str(input("Press any key to close the window"))
```